

## 0.1 MC Samples Used

The main MC samples used in the analysis are as follows,

- 200 M  $K \rightarrow \pi_D^0 \pi^+$
- 50 M  $K \rightarrow \pi_D^0 \mu^+ \nu_\mu$

## 0.2 Event Selection

- Require either the Control Trigger (sometimes described as Minimum Bias Trigger), Multi Track (MT) or Electron Multi Track (eMT) which are described in Chapter ??
- A vertex was selected if the total momentum was below  $78/\text{GeV}$ , a total positive charge of 1, a  $\chi^2$  less than 40 and a vertex position between 105 and 180 m.
- The tracks associated to the vertex are selected if they are in acceptance of all four Straw spectrometer chambers and the CHOD. The track momentum must be greater than  $5\text{ GeV}$  and less than  $60\text{ GeV}$  with a  $+$  of less than 30. The tracks must traverse within 20 cm of each other.
- Require the presence of a three track in the vertex with a selected positive charged pion ( $\pi^+$ ), and a selected electron positron pair ( $\pi^\pm$ ). The pion is selected using missing mass between the Kaon momentum which is taken from the average beam momentum and the two positive tracks. If the positive track is within 5 sigma of the  $\pi^0$  mass hypothesis then it is selected as a charged pion. If both positive tracks satisfy the condition the event is vetoed. From

MC only approximately 0.1/, % of Dalitz decays have both positive tracks satisfying the selection of the charged pion. The electron was assigned to the only negative track and the negative track which doesn't pass the charged pion selection was selected as the positron.

- A selected photon is also required. The missing momentum of the vertex was calculated with the Kaon momentum and the total vertex momentum and was used to extrapolate a straight line from the vertex position to the z plane of the LKr calorimeter. Within a radius of 15 cm of the extrapolated position a single cluster with a minimum energy of 5 *GeV* is selected as a photon from the decay. The photon energy has to satisfy the condition of being within 5 *GeV* of the calculated missing vertex momentum.
- An elliptical cut is used to reduce background that would otherwise pass the selection. The reconstructed mass of the electron, positron and photon is compared against the reconstructed mass of the electron, positron, photon and charged pion.

The cut on the reconstructed missing mass of the kaon, charge pion system was calculated using the Dalitz MC sample and is shown in Figure 1. The sigma is taken from the fitted guassian on the plot and calculated as,

$$\sigma = 3.984 \pm 0.007 \tag{0.1}$$

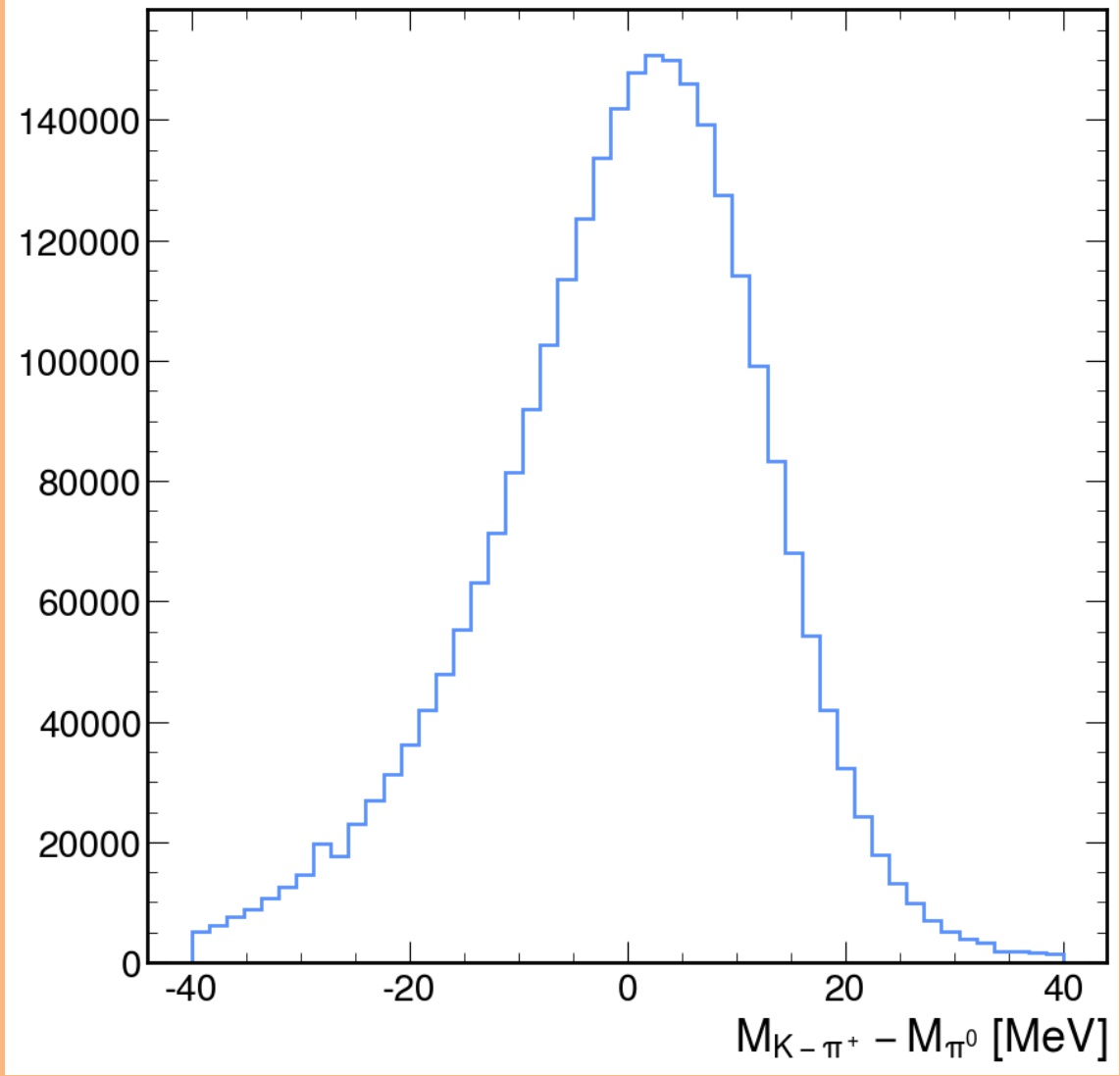


Figure 1: Reconstructed missing mass of Kaon and charged pion compared to the known neutral pion mass

The elliptical cut on the final reconstructed masses was defined using the Dalitz MC sample to collect the maximum amount of events passing the selection while making the cut as tight as possible. The sigmas for the reconstructed masses  $M_{ee\gamma}$  and  $M_{\pi ee\gamma}$  is calculated from fitting the distributions shown in Figure 2. The calculated sigma for the reconstructed mass  $M_{\pi ee\gamma}$  was calculated as

$$\sigma_{M_{\pi ee\gamma}} = 2.47 \pm 0.004 \quad (0.2)$$

and for  $M_{ee\gamma}$ ,

$$\sigma_{M_{ee\gamma}} = 1.329 \pm 0.009 \quad (0.3)$$

which were used as a origin for the elliptical parameters for the cut.

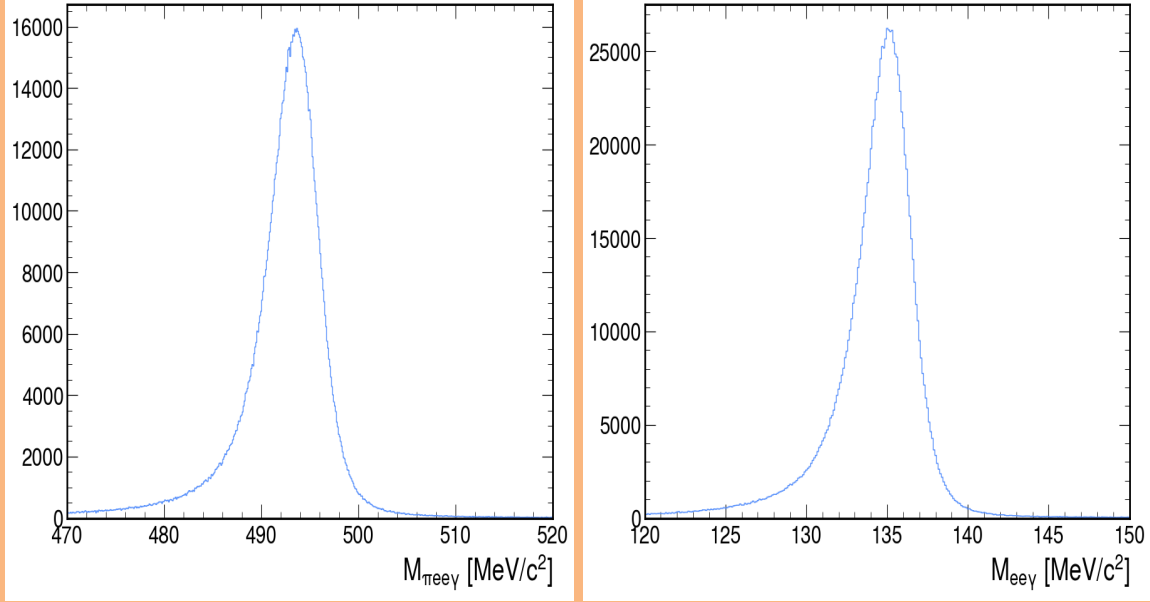


Figure 2: Reconstructed masses of selected particles, left the total of charged pion, electron, positron and gamma and on right, electron, positron and gamma.

The equation used for the elliptical cut in the selection is defined below,

$$\frac{((x - x_c)\cos(\theta) + (y - y_c)\sin(\theta))^2}{a^2} + \frac{((y - y_c)\cos(\theta) - (x - x_c)\sin(\theta))^2}{b^2} \leq 1 \quad (0.4)$$

where  $x_c$  and  $y_c$  is the centre of the ellipse in the x and y dimensions.  $\theta$  being the rotation of the ellipse,  $a$  and  $b$  are the length of the semi-major and the semi-minor axis respectively.  $x$  and  $y$  are the values from the selection and the event is selected if the equation returns a value that is less than or equal to 1. After first starting with the results from Figure 2 and optimizing for a tight cut with a acceptable number of events passing produced parameters below,

- ( $x_c = 493.37$ ): The x-coordinate of the ellipse's center.
- ( $y_c = 134.89$ ): The y-coordinate of the ellipse's center.

- ( $a = 10.24$ ): The semi-major axis of the ellipse ( $4 \times \sigma_{M_{\pi ee\gamma}}$ ).
- ( $b = 4.29$ ): The semi-minor axis of the ellipse ( $3 \times \sigma_{M_{ee\gamma}}$ ).
- ( $\theta = 0.4887$ ): The rotation angle of the ellipse in radians.
- ( $(x, y)$ ): The point being tested for inclusion within the ellipse.

Which produced an ellipse used in the selection which is shown in 3

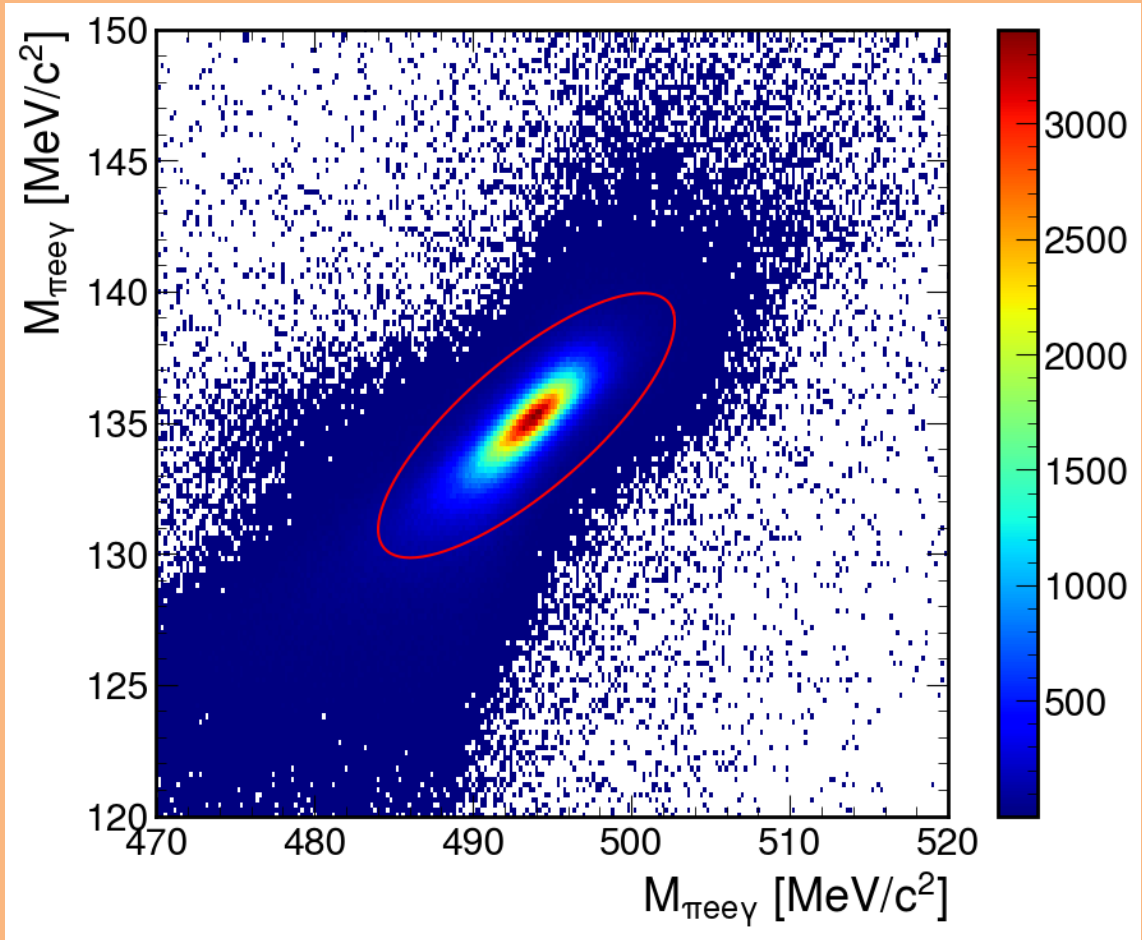


Figure 3: Reconstructed missing mass of  $M_{\pi ee\gamma}$  compared against  $M_{ee\gamma}$  produced from a K2pi Dalitz MC sample, with the defined elliptical cut used in the selection

The acceptance of the Dailtz sample is defined across the reconstructed electron position mass in the range  $2m_e < M_{ee} < m_{\pi^0}$  and is compared against the acceptance of the NA48/2 analysis [?] in Figure 4. Improving the selection using

kinematic cuts compared to the calorimeter PID improved the acceptance at high  $M_{ee}$  compared to NA48/2. It was expected that NA48/2 would have an acceptance advantage due to its design as a shorter experiment. If the same analysis procedure was employed using calorimetric PID the acceptance would not be competitive with NA48/2

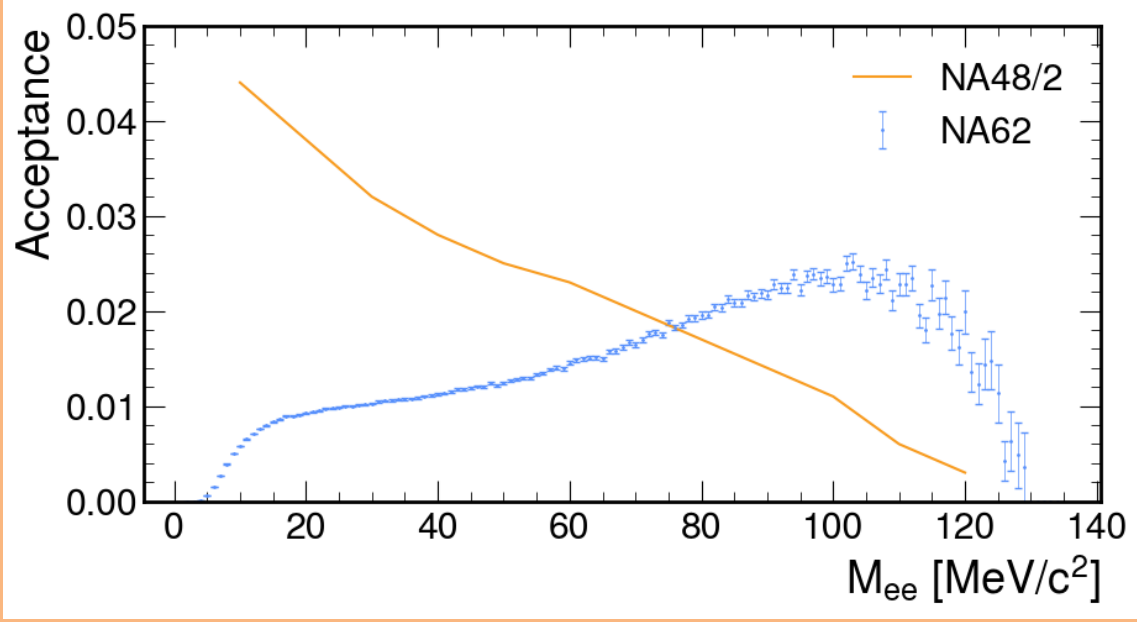


Figure 4: Acceptance of the NA62 selection compared against the acceptance produced in the NA48/2 analysis [?] ]

### 0.3 Integrated Kaon Flux

The total number of data events passing the selection for 2022 was 22,059,350 with the number of  $K^+$  decays within the fiducial volume for the 2022 dataset was computed as

$$N_K^{2022} = \frac{N_{selected}}{(A_{2\pi D}B_{2\pi D} + A_{\mu 3D}B_{\mu 3D})} = (1.65 \pm 0.05) \times 10^{12} \quad (0.5)$$

Where  $N_{selected}$  is the number of data events passing the selection,  $A_{2\pi D}$  and  $A_{\mu 3D}$  are the acceptances of the selection and are evaluated using MC samples and  $B_{2\pi D}$ ,

$B_{\mu 3D}$  are the branching fractions of the decays [? ]. As the MC samples employ the trigger emulators provided by the NA62 Framework the trigger efficiencies for both L1 and HLT are accounted for within the acceptances that are defined in Table ??.

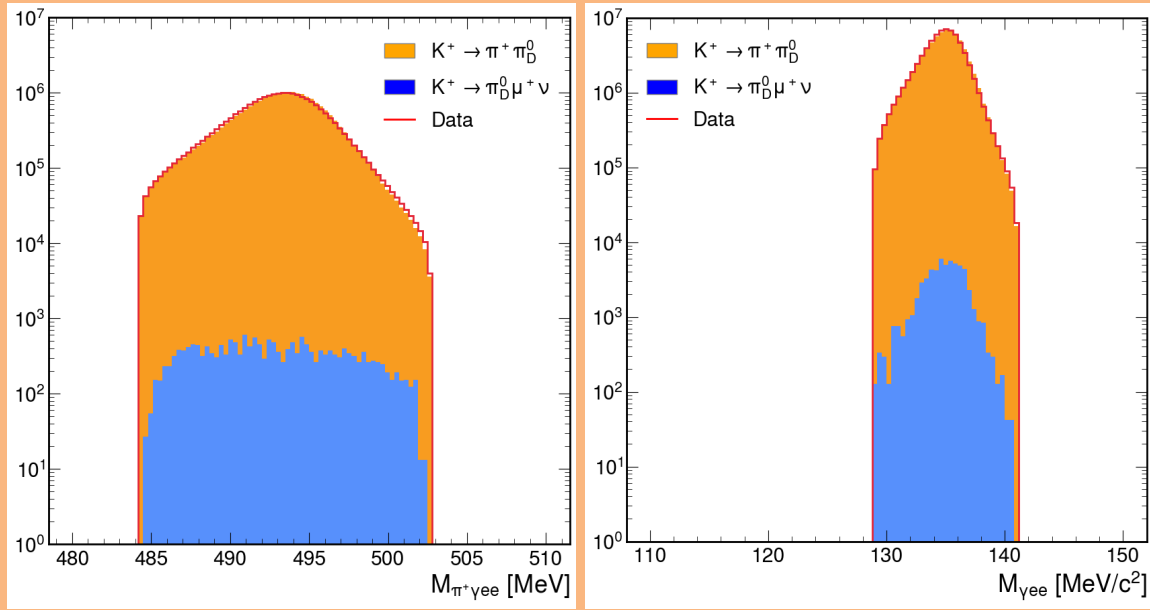
| MC Sample    | Acceptance |
|--------------|------------|
| $K_{2\pi D}$ | 0.56%      |
| $K_{\mu 3D}$ | 0.0027%    |

Table 1: Acceptances of  $K_{2\pi D}$  and  $K_{\mu 3D}$  samples for the  $K_{2\pi D}$  selection

The 2022 sample was available when the MC background method was employed. However when the second and final background method was finalised the 2023 dataset was available so it was used. The integrated number of Kaons was then calculated using the 66,839,882 data events passing the selection for 2022+2023

$$N_K^{2022/23} = (4.95 \pm 0.15) \times 10^{12} \quad (0.6)$$

The large increase in data events passing the selection when the 2023 dataset was added was due to the change in downscaling for the eMT trigger which went from  $3 \rightarrow 1$ . The reconstructed masses for data and MC passing the  $K_{2\pi D}$  selection for 2022+2023 are shown in Figure 5. The MC samples have been scaled using the calculated  $N_K$ . The plots for the 2022 sample produce similar results but with lower statistics.

Figure 5: Reconstructed masses  $K_{2\pi D}$  and  $K_{\mu 3D}$  for MC and data

## 0.4 Dark Photon Search

Two methods were employed in the background estimation for the Dark Photon Search. First used generated MC samples of the expected background as the background estimate, however as described below produced unsatisfactory results, which led on to the second method, using a data driven method for the background estimation. Both methods are described below

### 0.4.1 MC Background Method

As stated before, the MC method only used the 2022 dataset. It was expected that the limited size of the MC samples could impact the final result when used for the background estimate, however it was not feasible to directly increase the statistics on the main samples due to compute time and file size. It was possible to increase the statistics across a portion of the reconstructed mass spectra using a feature of

the Framework. It is possible for the true invariant mass of the electron positron pair to be used to calculate the kinematic variable  $x$  from Equation 0.7.

$$x = \frac{(Q_{e^-} + Q_{e^+})^2}{m_{\pi^0}^2} = \frac{m_{ee}^2}{m_{\pi^0}^2} \quad (0.7)$$

The resultant distribution from the MC sample is shown in Figure 6. The decay width in Equation 0.8 is strongly dependent on this kinematic term, reducing to zero as the reconstructed invariant mass reached the  $\pi^0$  mass. The Framework allows for the simulation of the Dalitz decay where the kinematic variable  $x > 0.5$ . From the true MC distribution in Figure 6 the fraction of events above 0.5 was 0.003. Producing a simulation with the enabled, even with a fraction of size of the main sample would decrease the errors on the MC counts by orders of magnitude.

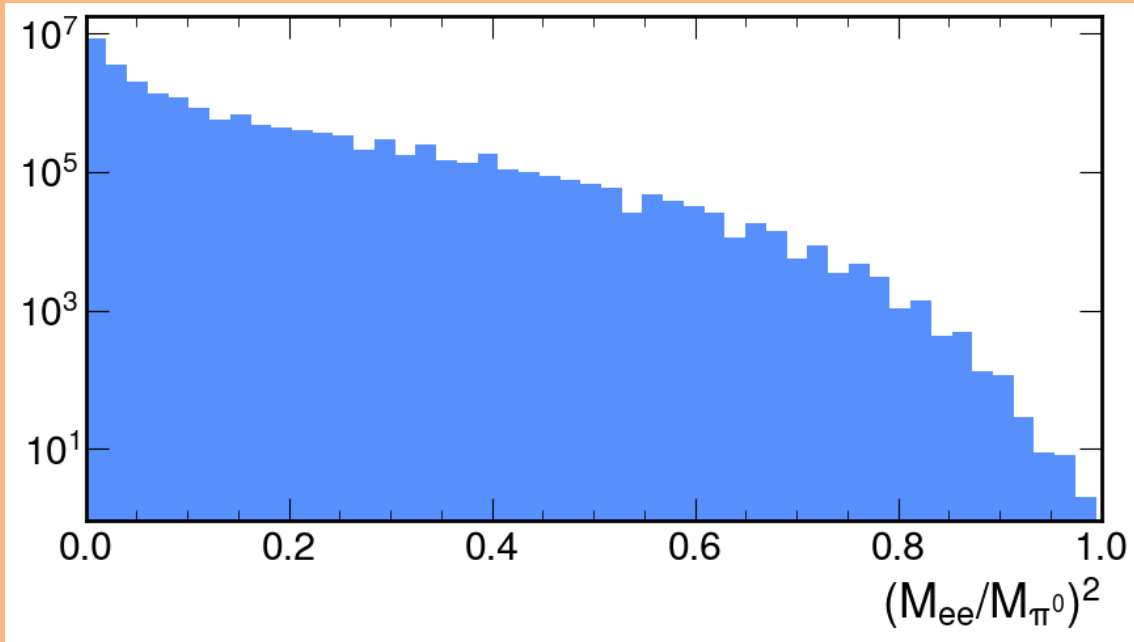


Figure 6: Reconstructed missing mass of electron positron pair of in terms of reconstructed kinematic variable  $x$

For the rest of the study the following MC samples were used

- 200 M  $K \rightarrow \pi_D^0 \pi^+$  ( $x < 0.5$ )

- 5 M  $K \rightarrow \pi_D^0 \pi^+ (x > 0.5)$
- 50 M  $K \rightarrow \pi_D^0 \mu^+ \nu_\mu$

In order to improve the combination of the two  $K_{2\pi_D}$  samples the large sample had events where  $x > 0.5$  removed to allow for the new sample to cover that region. After the full selection on the MC samples and data, the reconstructed mass of the electron positron pair is shown in Figure 7.

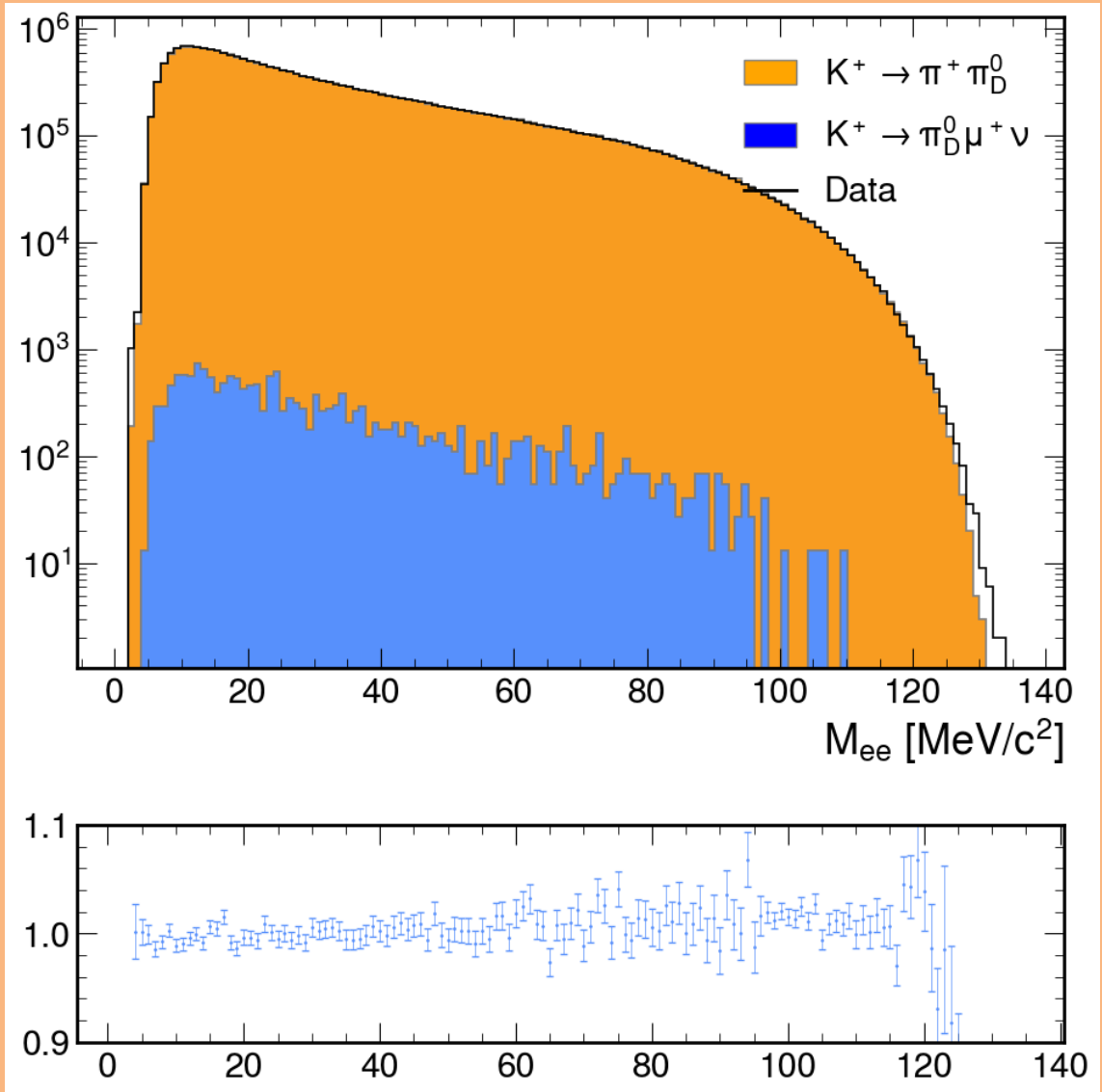


Figure 7: Reconstructed mass of electron positron pair of data and MC samples.

The samples used include the simulation of the  $\pi_D^0$  using the lowest-order differ-

ential decay rate [?] and is defined below

$$\frac{d^2\Gamma}{dxdy} = \Gamma_0 \frac{\alpha}{\pi} |F(x)|^2 \frac{(1-x)^3}{4x} \left( 1 + y^2 + \frac{r^2}{x} \right) \quad (0.8)$$

where  $\Gamma_0$  is the  $\pi^0 \rightarrow \gamma\gamma$  differential decay rate,  $\alpha$  is the fine-structure constant,  $F(x)$  is the pion transition form factor (TFF) and  $x, y$  are kinematic variables defined below

$$y = \frac{2Q_{\pi^+}(Q_{e^-} - Q_{e^+})}{m_{\pi^0}^2(1-x)} \quad (0.9)$$

where  $Q_{e^-}$ ,  $Q_{e^+}$  and  $Q_{\pi^+}$  are the four-momentum of the electron, positron and charged pion. The TFF is parametrised as  $F(x) = 1 + ax$ , with  $a$  being a parameter set by experiments or theory. ADD THEORY BIT. The Particle Data Group (PDG) average for all experiments was  $a = 0.03350.0031$  [?]. NEED MORE DESCRIPTION WHY CANT USE THEORY OR EXPERIMENT. The background estimation using the MC cannot rely on the input of theory or experiments due to the underlying uncertainties. In the NA62 Framework  $a = 0.032$  so a correction had to be made to remove the applied TFF and re-weight using the events passing the selection. It is clear that in Figure 8 which shows the data over MC from Figure 7 but rather than in terms of  $M_{ee}$  plot against  $x$  from Equation 0.7, suggests that there is a discrepancy. MC and data no longer match but is dependent along  $x$  suggesting that the TFF found in the Framework is not acceptable for use in the background estimate.

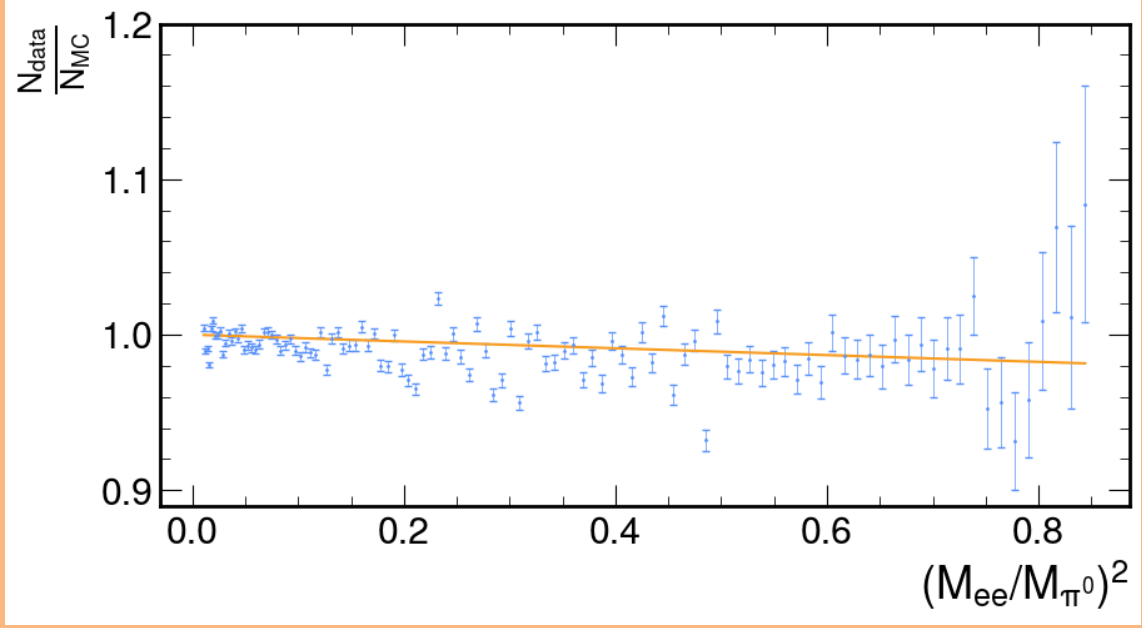


Figure 8: Data vs MC in terms of reconstructed kinematic variable  $x$  with the framework transition form factor

The effective TFF of the analysis can be found by removing the Framework value by inverting the effects of the square of the TFF,

$$F(x)^2 = (1 + ax)^2 \quad (0.10)$$

which resulted in the distribution shown in Figure 9. The MC with the removed effects of the TFF was fitted with equation 0.10 using a least  $\chi^2$  method, with the resultant line shown in Figure 9, representing a value for  $a$

$$a = 0.0210 \pm 0.0029 \quad (0.11)$$

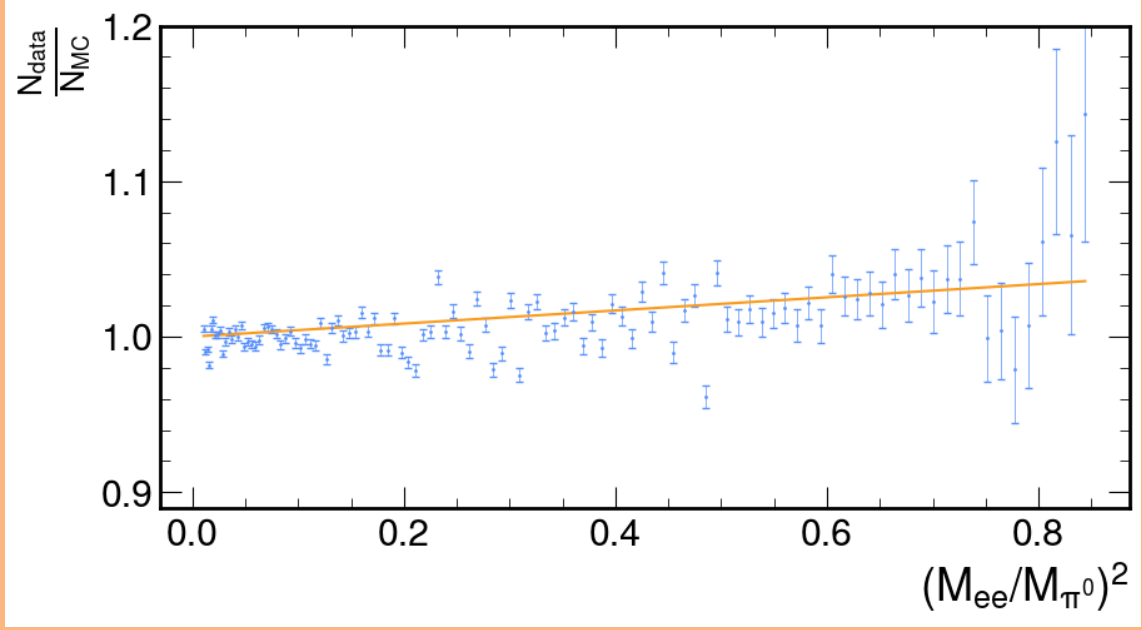


Figure 9: Data vs MC in terms of reconstructed kinematic variable  $x$  with the framework transition form factor removed

The correction resulted in slightly better Data/MC agreement shown in Figure 10 when compared to Figure 7.

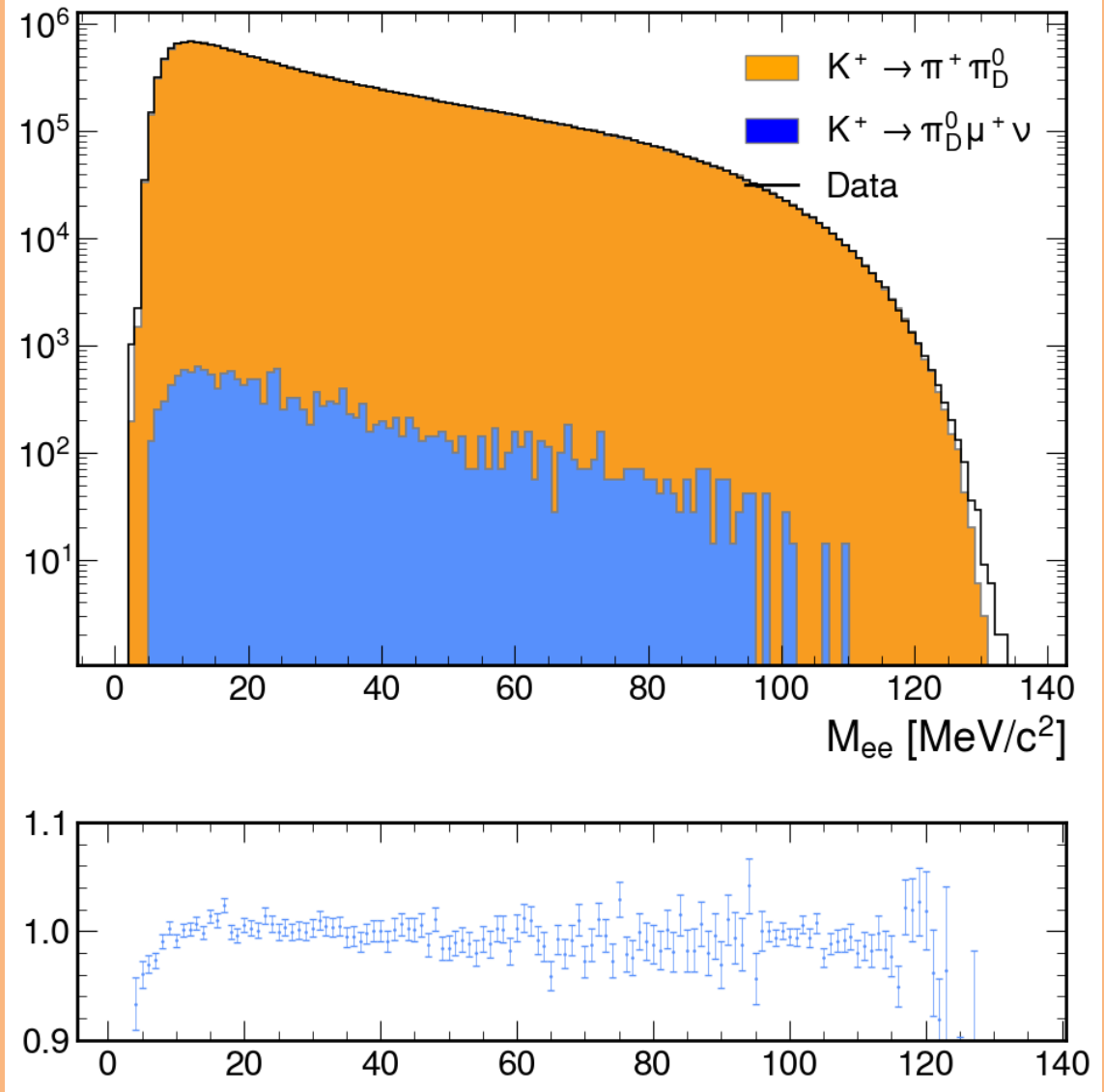


Figure 10: Reconstructed mass of electron positron pair of data and MC samples corrected for the effective transition form factor.

The final counts for events passing the selection in terms of the reconstructed invariant mass for each MC sample is shown in Figure 11, where it is clear where the two  $K_{2\pi_D}$  samples have been combined.

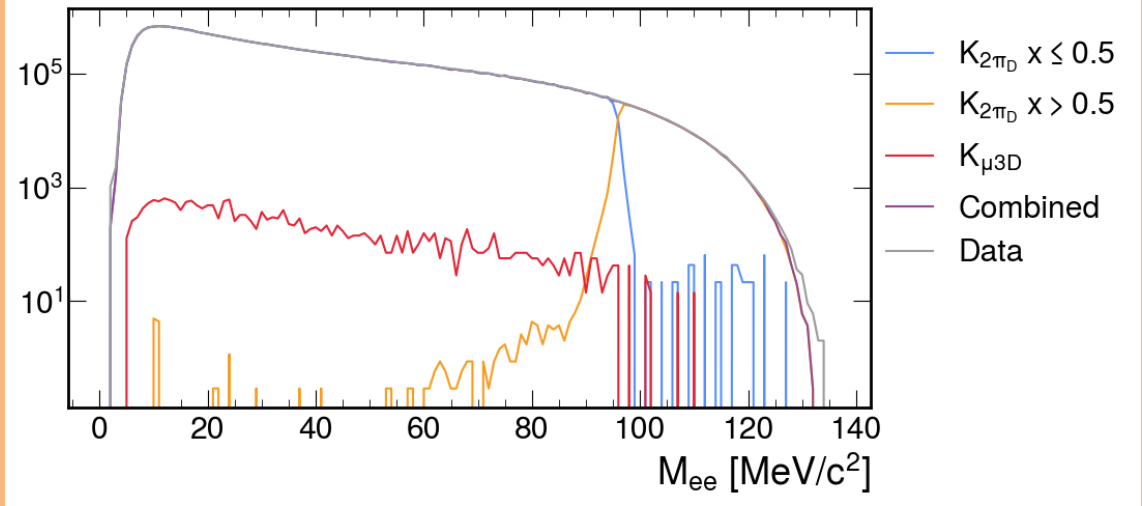


Figure 11: Reconstructed mass of electron positron pair of data, individual MC samples and combined MC.

The resultant statistical errors were calculated for each individual MC sample and combined to produce a combined MC error for each mass bin and is shown in Figure 12. It was clear that implementing the biased  $K_{2\pi_D}$ , significantly reduced the statistical error of the combined sample by an order of magnitude compared to using only the original sample..

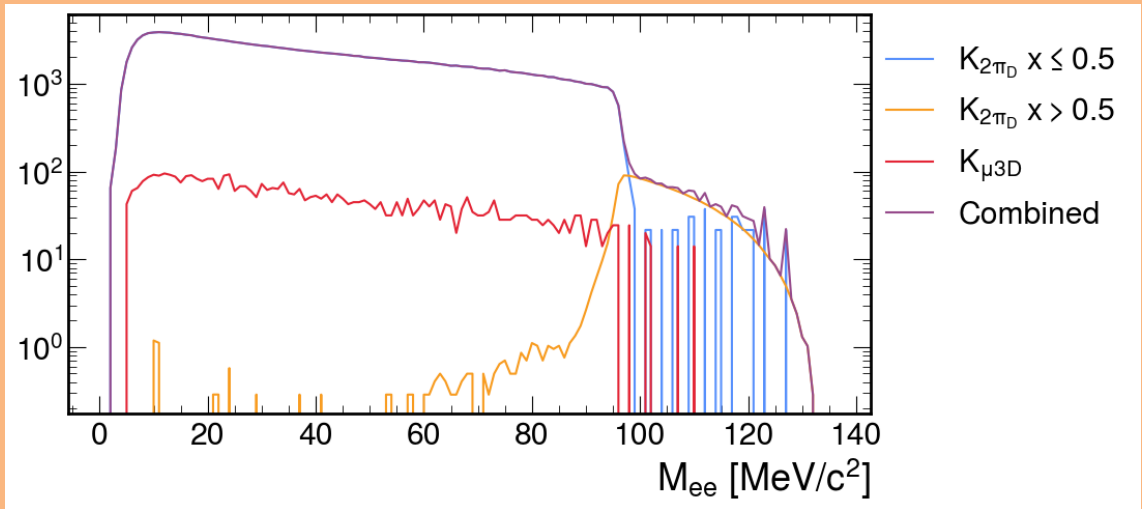


Figure 12: Statistical error of MC samples in terms of reconstructed mass of electron positron pair.

### 0.4.2 Data Driven Background Method